ML CLASSIFICATION PROJECT

1. OBJECTIVE

To build and evaluate supervised machine learning classification models using Python.

2. DATASET

The Breast Cancer Wisconsin dataset is obtained from the scikit-learn library. It contains features used to classify tumors as malignant or benign.

3. ALGORITHMS USED

1. Logistic Regression
2. Random Forest Classifier

4. METHODOLOGY

1. Loaded the dataset using scikit-learn.
2. Divided the dataset into 80% training data and 20% testing data.
3. Applied five-fold cross-validation.
4. Trained two classification models.
5. Evaluated the models using accuracy, precision, recall, F1 score, and ROC-AUC.
6. Compared the models using graphs and evaluation metrics.

5. EVALUATION METRICS

Accuracy: Measures the overall correctness of predictions.

Precision: Measures how many positive predictions are correct.

Recall: Measures how many actual positive cases are identified correctly.

F1 Score: Combines precision and recall into one score.

ROC-AUC: Measures how well the model distinguishes between the two classes.

6. RESULTS

The performance of Logistic Regression and Random Forest was compared using the evaluation metrics generated by the Python notebook. The model with the highest F1 score was selected as the best-performing model.

7. CONCLUSION

This project demonstrates the implementation of supervised machine learning classification using Python. Both algorithms were trained and evaluated, and their performance was compared using multiple evaluation metrics.

8. TOOLS AND TECHNOLOGIES

Programming Language: Python

Environment: Jupyter Notebook

Libraries: pandas, scikit-learn, matplotlib, seaborn

Dataset: Breast Cancer Wisconsin dataset